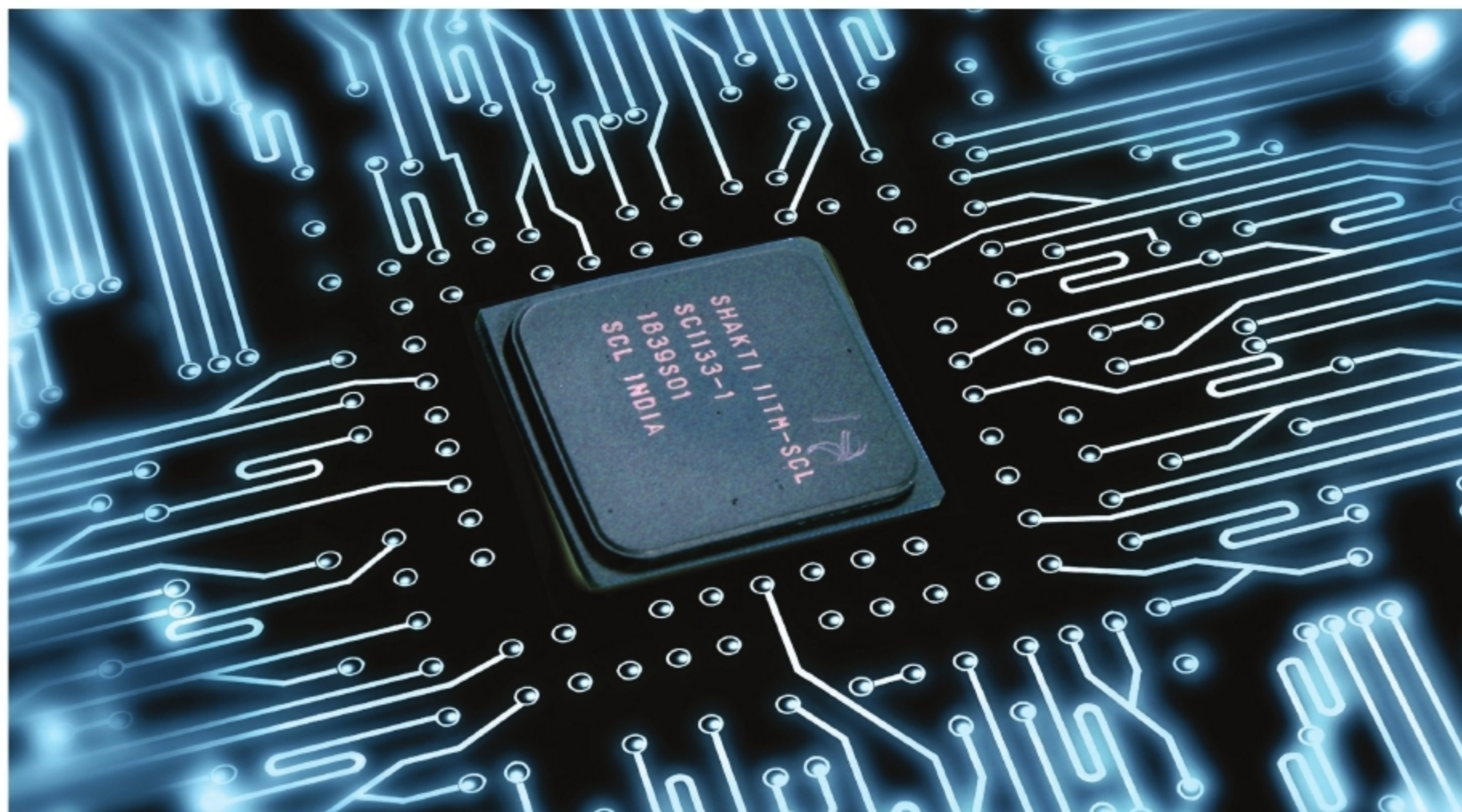


“The open source chip movement has just begun”

In October 2018, India saw the launch of the first indigenously built chip that can change the Indian consumer and industrial electronics landscape. Baishakhi Dutta catches up with **G.S. Madhusudan, senior project advisor at IIT-Madras**, to talk about the design strategies for SHAKTI microprocessor and how it can serve the electronics community at large.



Q What's the technology behind SHAKTI and its applications?

A. SHAKTI was started not just as a research project but to also develop production-ready processor cores, which are in competition with equivalent processors and are better in areas like security and functional safety. We will be competitive with equivalent ARM cores, better in some respects and lagging in others. We will know more once SHAKTI gets wider adoption.

The research part of SHAKTI focuses on areas like supercomputers, neuromorphic computing and security

techniques that use it as a base for research on publicly available platforms. This enables seamless translation into products and easier collaboration with other researchers.

SHAKTI aims to develop six families of processors, for everything from what goes inside door locks to supercomputers. For this kind of range, you cannot have a single design. In SHAKTI, E class is the lowest, then comes C class (for low-end cellphones, industrial controllers), I class, S class (for servers), H class (for supercomputers), F class (for functional safety processors for

aerospace/auto/medical applications) and finally, T class (for high-security applications).

Q How do you ensure that the chips are at par with international standards?

A. When we evaluate chips, there are basically three broad parameters: performance, power consumption and size. The instruction set is an international standard called RISC-V. The RISC-V consortium, of which IIT-M is a founding member, comprises experts from around the world, who have designed RISC-V ISA to be an